

Mid Semester Examination 2025

Name of the Program: B.Tech Petroleum Engineering
Name of the Course: Petroleum Refining Engineering

Semester: V
Course Code: TPE-505

Time: 1-1/2Hour

Maximum Marks: 50

Note:

- i. Answer **all the questions** by choosing **any one of the sub questions**.
- ii. Each question carries 10 marks

Q1	(10 marks)	
(a)	What is desaltation? What are the parameters on which crude oil characterize. Discuss with the definition?	CO 1, CO 2
	OR	
(b)	What is the goal of refining crude oil? Tabulate the Process Development Since The Commencement Of The Modern Refining Era	
Q2	(10 marks)	
(a)	Discuss the composition of crude oil? What are the major processes involved in petroleum refining. Discuss one of them	CO 2
	OR	
(b)	What is ADU? Draw the flow sheet diagram of (ADU) and discuss its process description with step wise?	
Q3	(10 marks)	
(a)	Define the following: i. API Gravity ii. Cloud point iii. Chemical dehydration iv. Penetration test v. Dynamic viscosity	CO 1
	OR	
(b)	Why and how desalting/dehydration of the crude oil can be done explain with step wise	
Q4	(10 marks)	
(a)	Differentiate between Naphtha's and Naphthalene? What are the methods of manufacture of naphtha's? Discuss composition and uses of naphthas.	CO 1, CO 2
	OR	
(b)	With the help of flow sheet diagram of simple crude oil distillation, Explain its process description	
Q5	(10 marks)	
(a)	What do you mean by blending of crude oil? Draw the flow sheet diagram of solvent dewaxing unit and explain its process description.	CO 1, CO 2
	OR	
(b)	Why we use MEK during solvent extraction? Explain the general steps in solvent dewaxing? What are the most widely used extraction solvent?	

